



**FACULTY OF COMPUTING**

**SEMESTER 1/20252026**

**SECJ 3553 - ARTIFICIAL INTELLIGENCE**

**SECTION 03**

**ASSIGNMENT 2**

State Space Search

**THEME:**

HEALTHCARE

**LECTURER:**

DR. NORSHAM BINTI IDRIS

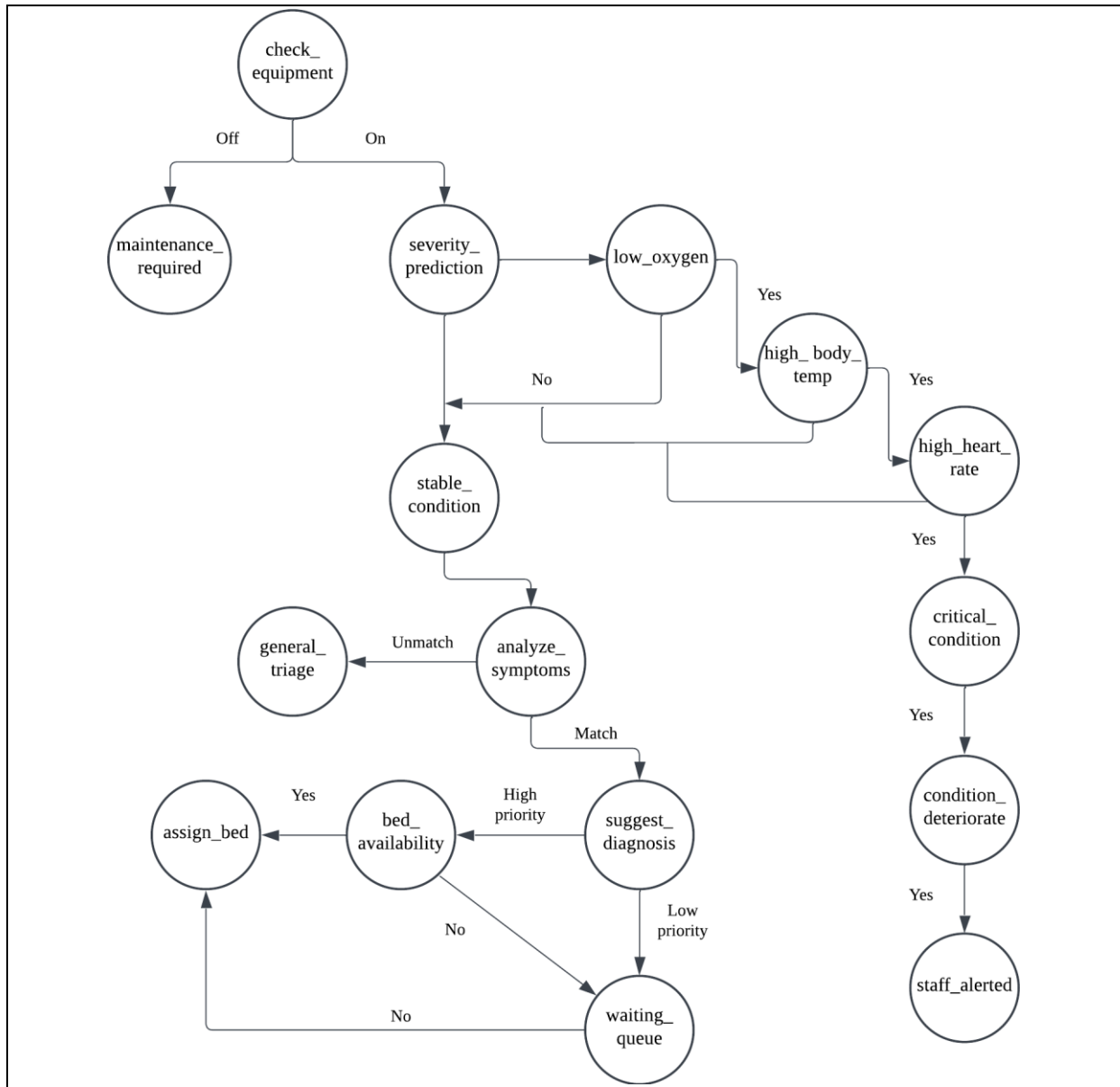
**Group no: Group 4**

| NAME                     | MATRIC ID |
|--------------------------|-----------|
| ELIJAH SHE YU SHENG      | A23CS0073 |
| LAU YEE WEN              | A23CS0099 |
| CHERYL CHEONG KAH VOON   | A23CS0060 |
| MUHAMMAD ADAM BIN RAZALI | A23CS0116 |

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## State Space Search



|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initial State = check_equipment</b>                                                                                                                                 |
| <b>Actions</b> = check_oxygen, check_body_temp, check_heart_rate, analyze_symptoms, predict_severity, check_bed_availability, trigger_staff_alert, request_maintenance |
| <b>Goal</b> = assign_bed OR staff_alerted                                                                                                                              |
| <b>Path Cost</b> = Yes, No, On, Off, Match, Unmatch, High priority, Low priority                                                                                       |

## Description

When the system checks the equipment, it will turn on if the medical device is functioning normally, otherwise it will switch off and request maintenance. Once the equipment is operating, the system will proceed to severity prediction. The system checks the patient's oxygen level, body temperature and heart rate. If all three values exceed the threshold, the patient is classified as critical. Otherwise, the patient is in a stable condition.

If the patient's vital are stable, the system will analyse the symptoms. When the symptoms match a known disease, the system will suggest a diagnosis, if they don't match, the patient is sent for general triage. The system then checks the bed availability for high priority patients. When a bed is available, it will assign the bed, otherwise, the patient will wait the queue.

If the patient's condition deteriorates become more critical, the system will trigger an alert to notify the staff.

## Peer review on Team Working (TW)

| Category                    | 4                                                                | 3                                                              | 2                                                                | 1                                                             |
|-----------------------------|------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Active participation</b> | Routinely provides useful ideas when participating in the group. | Usually provides useful ideas when participating in the group. | Sometimes provides useful ideas when participating in the group. | Rarely provides useful ideas when participating in the group. |

| MEMBERS                  | ELIJAH SHE YU SHENG | LAU YEE WEN | CHERYL CHEONG KAH VOON | MUHAMMAD ADAM BIN RAZALI |
|--------------------------|---------------------|-------------|------------------------|--------------------------|
| ELIJAH SHE YU SHENG      | -                   | 4           | 4                      | 4                        |
| LAU YEE WEN              | 4                   | -           | 4                      | 4                        |
| CHERYL CHEONG KAH VOON   | 4                   | 4           | -                      | 4                        |
| MUHAMMAD ADAM BIN RAZALI | 4                   | 4           | 4                      | -                        |